

A Structural Microsimulation of a Neighborhood Grocery Store: Theoretical Foundations, Model Architecture, and Design Rationale

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Abstract

This paper documents the theoretical foundations of a synthetic business data generator built to produce a three year history of a small neighborhood grocery store together with a fully known generating process. Rather than sampling numbers directly, the generator constructs a world: a street with locations, households with preferences and budgets, an owner with capital and beliefs, a calendar of seasons and shocks, and a daily market in which the two sides interact hour by hour. Every distributional choice in that world is defended by the same three question discipline, namely what the variable's support permits, what generative story produces it, and which distribution adds the least unwarranted structure beyond what is claimed. This paper consolidates that reasoning across the five construction phases, from the opening decision solved as a mixed integer program through the three year panel of customer turnover and capital accumulation, and explains why each structural choice was made rather than merely what it computes. The result is intended as a companion reference for readers who want to understand the model itself, independent of the software that implements it.

1 Introduction

Every real business dataset is the residue of a world. Behind a spreadsheet of receipts and invoices sit people with habits and budgets, a street with rents and foot traffic, and an owner with limited capital and imperfect information, all colliding day after day until the paperwork accumulates. A dataset built to teach or test data analysis honestly should therefore not be built by drawing numbers from convenient distributions and calling the result a business. It should be built the other way around: construct the world first, with its people, its constraints, and its causal structure, and let the paperwork fall out of the world's own operation.

This paper documents the theoretical reasoning behind such a construction. The subject is a simulated small grocery store, run by a single owner over three years, generating the full range of records a real shop produces: till receipts, supplier invoices, shelf price histories, promotional campaigns, monthly ledgers, and a tax filing. What distinguishes the dataset from an arbitrary spreadsheet of plausible numbers is that every number has a traceable cause. Prices respond to costs, costs respond to shocks, shoppers respond to prices according to individually varying tastes, and the owner responds to what he observes, which is never the same as what is true. The gap between what the owner believes, what actually happens, and what a perfectly informed operator could have achieved is not injected noise; it is the mechanical consequence of a chain of modeling decisions, each of which this paper defends on its own terms.

The construction proceeds in five phases, summarized in Figure 1 and detailed in the sections that follow. The first phase builds the world as it exists the moment before the store opens: a street of candidate locations, a population of households with preferences, and an owner who must decide where to open, what to stock, and how much capital to commit, all before a single customer has walked through a door. The second phase writes the script of the year that follows, meaning every exogenous influence on the world, such as the weather, the calendar, and macroeconomic shocks, decided in advance and independent of anything the simulated agents do. The third phase performs that script: customers and the owner interact hour by hour, and every transaction, stockout, and promotional decision is recorded. The fourth phase turns the single performed year into a laboratory, allowing the same world to be replayed under edited conditions so that the causal effect of a tax reform or a competitive shock can be measured exactly rather than estimated. The fifth phase extends the horizon to three years, adding customer turnover, capital accumulation, and a sequence of scripted events that a real small business might plausibly encounter.

The remainder of this paper is organized around that same five phase structure. Section 2 positions this construction against the other ways synthetic and case study business data are currently produced, and discusses what the resulting dataset makes possible, both analytically and pedagogically. Section 3 states the design principles and notational conventions that recur throughout the model. Sections 4 through 8 treat each phase in turn, explaining not only how

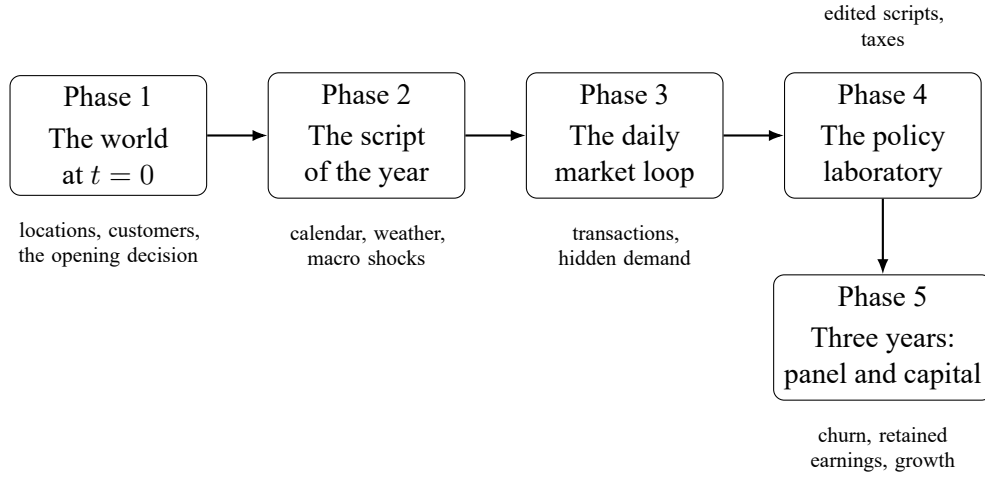


Figure 1: *The five phase construction. Each phase consumes the outputs of the ones before it and adds one new kind of structure to the world.*

each mechanism is built but why it was built that way rather than some simpler or more elaborate alternative. Section 9 describes the accounting layer that ties every euro and every unit of stock to a specific artifact, including the deliberate imperfections that make the raw records disagree until they are cleaned. Section 10 reflects on the architecture as a whole, and Section 11 closes the paper.

2 Related Approaches and the Case for Structural Synthesis

Synthetic business data is not a new idea, and it is worth being precise about where this construction sits relative to the approaches already in routine use, rather than presenting it as though it emerged in a vacuum. Four traditions dominate the current landscape, each built to optimize for something other than what this project optimizes for.

2.1 Rule based generators

The simplest and most widely used tools, such as Faker and Mockaroo, fill a schema with plausible looking field values: names, addresses, dates, amounts drawn from simple parametric distributions attached independently to each column. These tools are fast, require no seed data, and are entirely adequate for their actual purpose, which is populating a database or testing a software pipeline against realistically shaped inputs. They are not built to support analysis practice, because each column is generated independently of every other: there is no mechanism linking a customer’s income to their basket size, no mechanism linking weather to foot traffic, and no sense in which the resulting numbers were caused by anything. A student who regresses one generated column on another is not discovering a relationship, because none was ever built in; every correlation Section 4 through Section 8 of this paper painstakingly justifies is, in a rule

based generator, simply absent by construction.

2.2 Statistical and generative model synthesizers

A more sophisticated family of tools learns to imitate the statistical shape of a real dataset rather than sampling columns independently. Conditional generative adversarial networks such as CTGAN, variational autoencoders, and copula based tools bundled in synthesis libraries such as the Synthetic Data Vault (Patki, Wedge, and Veeramachaneni, 2016; Xu, Skoularidou, Cuesta-Infante, and Veeramachaneni, 2019) are trained on a real seed dataset and learn to reproduce its marginal distributions and pairwise correlations closely enough to substitute for it in downstream use, often motivated by a differential privacy guarantee (Dwork, 2006) that lets an organization share a synthetic look-alike without exposing real customer records. This is a genuine and valuable technology for its actual purpose, which is privacy preserving data sharing. It is a poor fit for teaching data analysis for three specific reasons. First, it requires an existing real dataset to imitate, so it cannot manufacture a genuinely new case; every synthetic version is a shadow of something that already happened elsewhere. Second, the generator itself is a black box: a neural network or a fitted copula reproduces a distribution's shape without asserting any mechanism that produced it, so there is no known ground truth against which a student's causal claim, elasticity estimate, or forecast can be graded, only the observed data the student was already given. Third, and most importantly for this project's purposes, a distribution matching model has no way to answer a counterfactual question, because it was never given a reason for any number, only a shape to reproduce; asking such a model what revenue would have been without a tax change is not merely hard, it is category confusion, since the model has no representation of a tax change to remove in the first place.

2.3 The business case method

Business schools have long taught with hand authored cases: a narrative description of a company's situation accompanied by a set of exhibits, built by a case writer around a specific pedagogical point. This tradition, associated most closely with Harvard Business School's case method, remains the gold standard for teaching judgment under ambiguity, and the case narrative structure adopted for this project's own teaching materials borrows directly from it. Its limitation is not narrative quality but process: each case is bespoke, manually tuned by its author to make the numbers come out right, produced once, and then reused unchanged for years. There is no underlying generative process to interrogate, no way to produce a counterfactual twin of the same company under a different tax regime, and no way to grade a student's causal claim against a documented mechanism, because the case writer's spreadsheet is the only ground truth that exists and it was authored to support a conclusion rather than to allow one to be discovered.

2.4 Structural microsimulation

The methodological tradition closest to this project's own is structural microsimulation, founded by Orcutt's (1957) proposal to model an economy as a population of individual units following explicit behavioral rules rather than as aggregate equations, and carried forward today in tax and benefit microsimulation models such as EUROMOD and in agent based market and transport simulations across applied economics. These models share this project's central commitment: agents obey genuine, documented behavioral rules, so a policy change can be replayed through an unchanged population and the difference is a true causal effect rather than an estimated one, exactly the property Section 7 builds into this model's own policy laboratory. Where such models typically diverge from this project is in what they are built to produce. Their output is usually an aggregated statistic destined for a policy briefing, such as a national poverty rate or a revenue projection, rather than a transaction level business record; and because their audience is professional economists rather than students learning to clean and reconcile data, they rarely simulate the imperfect paperwork of Section 6, since a policy model has no reason to misfile an invoice. Structural microsimulation gets the causal architecture right; it has simply never been aimed at the specific problem of teaching an individual analyst to reconstruct a single business's truth from its own imperfect records.

2.5 What this construction combines

No single tradition surveyed above combines all four of the properties this project requires simultaneously. It requires, first, a fully documented structural generating process rather than a black box, so that every pattern in the data has a named cause, in the spirit of structural microsimulation rather than distributional synthesis. It requires, second, transaction level business artifacts, till receipts and supplier invoices rather than aggregated statistics, carrying deliberately realistic and separately documented paperwork imperfections, in the spirit of a genuine business record rather than a clean simulation output or a case writer's tidy exhibit. It requires, third, exact counterfactual replay at the level of individual decisions, made possible only by the keyed randomness discipline of Section 3, which neither a hand authored case nor a distributional synthesizer can offer at all. It requires, fourth, a built in scorecard for the analysis itself: the three profit figures of Section 6 turn the abstract idea of analytical value into a specific number, the gap between what the owner believed, what he actually earned, and what a perfectly informed operator could have earned, which is not a feature any of the four traditions above were ever designed to provide, because none of them stage a protagonist who is wrong in a way that can be measured.

2.6 The potential of the resulting data

A dataset built this way supports a genuine progression of analytical questions rather than a single fixed exercise, because the same underlying mechanisms answer descriptive, diagnostic, predictive, prescriptive, and counterfactual questions alike, each gradeable against the hidden truth the generator itself recorded. At the descriptive layer, a class can ask where the money comes from and goes, when people shop, and what a basket looks like, all directly readable from the ledger and the receipts. At the diagnostic layer, the same data supports genuine causal questions with known answers: whether weather actually moves revenue once the calendar is controlled for, exactly the mechanism of Section 5; how much of a wholesale cost shock reaches the shelf and how quickly, measurable against the menu cost rule of the same section; and whether a price increase loses customers outright or merely redirects them to a close substitute, answerable against the discrete choice structure of Section 4. At the predictive layer, a forecasting exercise can be benchmarked not only against a held out quarter but against the owner's own documented forecasting rule, exposing exactly where and why a naive trailing average underperforms, per Section 6. At the prescriptive layer, a class can redesign the ordering policy or the markdown trigger and price the redesign in real euros against the oracle figure of Section 6, rather than merely arguing that a better policy would probably help. At the counterfactual layer, the scenario twins of Section 7 and the three year arc of Section 8 let a class ask what a tax cut actually did to prices, what a discounting competitor actually cost, and whether an expansion decision paid for itself, each answerable as an exact difference between two runs of the same world rather than an inference from a single history. No other synthetic data source surveyed above supports this particular ladder from description through causal counterfactual with a documented answer waiting at every rung.

2.7 Value for teaching scientific reasoning, not only data analysis

The deeper claim behind this construction is that its principal teaching value is not technical fluency with a particular software tool but a disciplined scientific habit of mind, and that a dataset with a fully known generating process is uniquely suited to teaching that habit because it can tell a student, definitively, when they are right. An analyst working this data is repeatedly placed in a position where a plausible looking pattern is a trap rather than a finding: the flat category of Section 5 whose apparent seasonality is a budget mediation effect rather than a genuine preference, the markdown campaign of Section 6 whose naive before and after comparison is contaminated by the very selection rule that triggered it, and the energy crisis of Section 5 whose use as a price instrument is subtly invalid because it also moves household budgets. Each of these traps has a documented, correct resolution, which means an instructor can teach the general scientific skill of interrogating one's own inference, not by asserting that skepticism is a virtue in the abstract, but by handing a student a specific claim, a specific method for checking it, and a specific

ground truth that reveals whether the checking actually worked. The same property extends beyond individual traps to the whole discipline of hypothesis formation: because the model is built from named mechanisms defended by the three question discipline of Section 3 rather than fitted after the fact to match a target, a student can be asked to propose a mechanism before seeing whether the data supports it, and then be told, precisely, whether they were right, which is the same discipline a working analyst must apply to a real business where no such answer key will ever be handed over. This is, ultimately, the justification for building the world first and letting the paperwork fall out of its own operation, as this paper's introduction proposed: only a dataset built that way can teach an analyst not merely how to compute an estimate, but how to know whether an estimate deserves to be believed.

3 Design Principles and Notation

3.1 The three question discipline

Every stochastic element of the model, from a customer's weekly grocery budget to the day a storm arrives, was chosen by answering the same three questions, always in the same order.

1. **Support.** What values can the variable actually take? A probability must live on $(0, 1)$, a count must be a nonnegative integer, a share of a composite basket must live on a simplex, and a cost must be strictly positive. The distribution chosen must respect this support without needing an artificial truncation.
2. **Generative story.** What process actually produces the variable? A quantity that arises from many small effects adding together behaves differently from one that arises from many small factors multiplying together, and both differ again from a quantity that arises from a fixed number of independent yes or no trials at a common rate.
3. **Maximum entropy.** Once the support and the generative story have narrowed the field, which distribution within that field adds the least unwarranted structure beyond the facts one is actually willing to claim? This is the maximum entropy principle in its practical form: if the only facts asserted about a variable are its mean and its variance, the distribution that respects exactly those two facts and assumes nothing else is the least presumptuous choice available.

This discipline explains a recurring pattern in the model. Wherever a quantity is the product of many small positive multiplicative factors, such as a household's grocery budget (itself a product of household size, income level, and consumption habits), the natural distribution is the LogNormal, because summing the logarithms of many small factors is exactly the setting in which the central limit theorem produces a Normal distribution, and exponentiating a Normal produces a LogNormal that is guaranteed positive. Wherever a quantity is a probability,

meaning it must live strictly between zero and one, the Beta distribution is the natural family, because it is the maximum entropy distribution on that support once a shape is fixed, and its two parameters give direct control over where the mass concentrates. Wherever a quantity counts independent rare events at a common rate, the Poisson distribution follows, because it is the maximum entropy distribution for counts given only a rate. These are not arbitrary choices repeated by habit; each one answers the same three questions afresh in its own context, and each answer is defended at the point where the variable first appears.

3.2 Notation

The model follows a McElreath style notation throughout, meaning every distribution is written with its natural parameters, and any scale parameter is a standard deviation rather than a variance, so that a Normal distribution is always written $\text{Normal}(\mu, \sigma)$ and never $\text{Normal}(\mu, \sigma^2)$. Three kinds of assignment recur.

1. The symbol $\sim$ denotes a stochastic draw, as in $X \sim \text{Beta}(2, 2)$, meaning X is a random realization from that distribution.
2. The symbol $=$ denotes a deterministic definition, as in $D_{cl} = \text{Households}_l \cdot \mu_c$, meaning the left hand side is computed exactly from the right hand side with no additional randomness.
3. The symbol $\stackrel{\leftarrow}{=}$ is used specifically where the direction of causation is itself the point being made, most often in the cost equations, to mark that the right hand side causes the left hand side rather than merely equaling it algebraically.

A hat, written $\hat{\cdot}$, always marks a belief held by the owner, as distinct from the corresponding true value that the owner cannot observe. Indices recur across the whole model: locations are indexed by l , categories of merchandise by c (there are twelve, matching a real product catalog), individual products or stock keeping units by s , customers by i , and time by t , measured in days across a horizon that eventually spans three years, with weeks indexed by w , months by m , and individual shock events by k . A small number of letters, such as α and m , are reused across different phases for unrelated quantities, exactly as they are reused in the underlying source material; every such reuse is flagged in prose at the point where it recurs, and the table below lists each meaning separately under the section where it first appears.

Every symbol used in a displayed formula anywhere in this paper is defined in prose at the point it first appears. Because the paper is long and a symbol's definition can be several pages before a later formula that reuses it, Table 1 collects every symbol in one place, organized by the section in which it is first introduced, purely as a reference for checking a formula against its plain language meaning without having to search back through the text.

Table 1: Notation used throughout the paper, grouped by the section in which each symbol first appears.

Symbol	Meaning	First used
<i>Indices and meta-notation</i>		
l, c, s, i, t	location, category, product/SKU, customer, and day indices	§3
w, m, k	week, month, and shock-event indices	§5–8
$\hat{\cdot}$	a belief held by the owner, as opposed to the true value	§3
$\sim, =, \stackrel{\leftarrow}{=}$	stochastic draw; deterministic definition; causal definition	§3
<i>Phase One — locations and the opening decision</i>		
Q_l	latent quality score of location l , on $[0, 1]$	§4
$\text{Households}_l,$ $\text{Rent}_l,$ SetupCost_l	a location’s catchment size, monthly rent, and fit-out cost	§4
N_l	pool of households at location l who adopt the store	§4
B_i	customer i ’s baseline weekly grocery budget	§4
d_i, π_i, p_i	primary shopping day; probability of shopping on it; probability of an extra midweek visit	§4
w_i, α	basket composition across categories; its Dirichlet concentration vector	§4
β_i	customer i ’s price sensitivity	§4
b_i, z_i, ν_i	brand affinity; standardized log budget; independent brand-affinity noise	§4
$T_i,$ PaysByCard_{iv}	card-preferring type; whether visit v is paid by card	§4
U_{is}	customer i ’s utility for product s	§4
θ_{ic}	customer i ’s current need for category c	§4
a_s, γ	product s ’s unmodeled appeal intercept; weight on brand fit	§4
BrandLevel_s	product s ’s position on the brand spectrum	§4
D_{cl}, μ_c	true demand potential for category c at location l ; mean monthly per-household consumption of category c	§4
$\hat{D}_{cl}, \delta_c$	owner’s believed demand; his log-scale belief error	§4
m_c	category c ’s fixed retail markup	§4
η, ρ	safety-stock fraction (0.30); cap on one product’s share of category demand (0.05)	§4
$\text{TotalCost}_0,$ $\text{FixedCost}_0,$ VariableCost_0	opening cost components, decomposed causally	§4
y_l, x_s	binary location choice; binary listing indicator	§4
q_s, u_s, E_0^h	units purchased at opening; believed units sold; extra staff hired	§4
<i>Phase Two — calendar, weather, and cost shocks</i>		
$T_t, \mu_T(t), \varepsilon_t,$ ϕ	day t ’s temperature; its seasonal cycle; its persistent anomaly; the anomaly’s AR(1) persistence (0.7)	§5

Symbol	Meaning	First used
W_t, p_{01}, p_{11}	whether it rains on day t ; dry-to-wet and wet-to-wet transition probabilities	§5
R_t	rainfall amount on a wet day	§5
M_{ct}	demand modifier for category c on day t	§5
a_c, κ_c, h_c	category c 's seasonal amplitude; weather-anomaly loading; pre-holiday loading	§5
z_t, H_t	standardized temperature anomaly; pre-holiday flag	§5
ψ_{pt}, a_p, κ_p	product-type p 's own seasonal tilt, amplitude, and anomaly loading	§5
Λ_t, ϕ^Λ	daily foot-traffic modifier; its AR(1) persistence (0.85)	§5
B_{iw}, ξ_{iw}	customer i 's week- w budget; its small weekly wobble	§5
s_{iw}, g_{iw}	tight-spell multiplier; splurge multiplier, both on B_{iw}	§5
$\bar{g}_t$ (or $\bar{g}_w$)	the scripted energy crisis's own trajectory over time	§5
$\text{infl}(t)$	the continuous annual inflation drift	§5
K, τ_k	number of shock events in the year; arrival day of event k	§5
ζ_k, ρ_k	peak magnitude of shock k ; its exponential decay time constant	§5
$g_k(t), A_k$	shock k 's effect on day t ; the set of categories it touches	§5
CostMult_{ct}	wholesale cost multiplier for category c on day t	§5
$\overline{\text{Cost}}_{s,t}, \alpha$	owner's smoothed invoice-cost trend for product s ; its EWMA weight (0.35, unrelated to the basket α above)	§5
<i>Phase Three — the daily market and the recording layer</i>		
P_{ict}, r_{ict}, r_{ic}	customer i 's pantry stock of category c ; its daily drain rate; the drain rate's undistorted base level	§6
$\theta_{ict}, P_{ic}^{\text{tgt}}, \tau$	need intensity from pantry shortfall; target pantry level; target days of cover (10)	§6
α (need shape)	shape constant of the pantry-shortfall need function, unrelated to the two earlier uses of α	§6
κ (purchase)	random fraction of the shortfall actually purchased, Uniform(0.7, 1.0)	§6
$\hat{D}_{cw}^{\text{op}}$	owner's trailing four-week sales forecast for category c	§6
Target_{sw}, R, L	order-up-to target for product s ; review period (7 d); lead time (2 d)	§6
$\tilde{\sigma}_{sw}$	product s 's smoothed share of its category's recent sales	§6
$\text{Spoiled}_{st}, \lambda_c, \lambda_c^d$	units of product s spoiled overnight; category's weekly base spoilage rate; its converted nightly rate	§6
$f_{ct}, \kappa_c^T, \kappa_c^E, \bar{T}$	spoilage factor; heat-term loading; cold-chain loading; annual mean temperature	§6
$D_{st}, B_{st}, S_{st}, A_{sc}$	book-minus-true discrepancy; book stock; true stock; monthly count correction	§6
$\pi_{\text{believed}}, \pi_{\text{realized}}, \pi_{\text{oracle}}$	owner's believed profit; the ledger's realized profit; a perfectly informed operator's oracle profit	§6
$\Delta_{\text{optimism}}, \Delta_{\text{analytics}}$	cost of the owner's overoptimism; value of better information and decisions	§6
<i>Phase Four — the policy laboratory and the tax layer</i>		
$r_c(t)$	category c 's value added tax rate path	§7

Symbol	Meaning	First used
$\text{Tag}_{st},$	product s 's shelf price tag; its smoothed tax-exclusive invoice cost	§7
$\text{NetCost}_{s,t}$		
VAT_t	net value added tax remitted in month t	§7
r^{pay}, r^{π}	payroll tax rate (0.25); profit tax rate (0.20)	§7
<i>Phase Five — the three year panel and capital</i>		
type_i, h	customer i 's persistence type (rooted or transient); transient customer's monthly departure hazard (1/18)	§8
ρ_{RE}, π_m	retained-earnings share (0.5); month m 's after-tax profit	§8
$\text{RE}_m, \text{capex}_m$	cumulative retained earnings through month m ; capital spent that month	§8
$K_{\text{exp}}, C_{\text{exp}}$	retained-earnings threshold that triggers expansion; one-time fit-out cost	§8
$m_i, \tilde{s}_i$	discounter's visit-probability penalty on customer i (unrelated to the markup m_c); customer's price-sensitivity percentile	§8
χ, k_i	competitor calibration constant; rooted/transient penalty weight	§8
<i>The accounting layer</i>		
$\text{book}(t),$ $\text{deliveries}(t)$	the book-inventory identity's stock and delivery terms	§9
$\text{sales}(t),$ $\text{write_offs}(t)$	the book-inventory identity's sales and write-off terms	§9
$\text{ProfitBeforeTax},$ $\text{ProfitTax}, \text{ProfitAfterTax}$	the annual profit-and-tax identity's three terms	§9

3.3 Reproducibility and the discipline of keyed randomness

The entire model is driven by a single seeded pseudorandom generator, so that the whole three year history can be reproduced exactly from one integer. This alone would be unremarkable were it not for a second requirement that follows from the model's purpose as a policy laboratory (Section 7): the ability to replay the same world under an edited script, such as one in which a particular shock never occurred, and to attribute every difference between the two runs entirely to the edit.

This requirement imposes a strict discipline on how randomness is consumed. If every random draw were taken from one sequential stream, then deleting a single event partway through the year would shift every subsequent draw, and the comparison between the two runs would be contaminated by reshuffled randomness that has nothing to do with the edit itself. The model instead keys every draw by the stable identity of the entity it concerns, such as a specific customer on a specific day, or a specific product on a specific night, using independently spawned child streams. A customer's Tuesday decision is drawn from a stream indexed by that customer and that day, so that in any counterfactual replay which does not touch that customer, the identical

draw is reproduced. This is the common random numbers technique familiar from the simulation literature, and it is adopted from the first line of the implementation rather than retrofitted later, because retrofitting it after the fact would require rewriting every random draw in the model.

4 Phase One: The World at Rest

Phase One constructs the world as it exists at the instant before the store opens. It proceeds from both sides of the market at once. The demand side begins with a street, populates it with households, and turns households into customers with individually varying tastes, budgets, and shopping schedules. The supply side begins with an entrepreneur holding a fixed sum of capital and forming beliefs about a demand he cannot directly observe, and ends with his first and largest decision: where to open, what to stock, how much of it to buy, and how many people to hire, all made at once because each choice constrains the others through the shared limits of capital and shelf space.

4.1 The street and its locations

Before there are customers, there is geography. The neighborhood offers a small number of candidate premises, and they are not equal. Every observable feature of a location, its rent, the number of households in its catchment, its fit out cost, and its physical capacity, is modeled as flowing causally from a single latent quality score $Q_l \in [0, 1]$:

$$\begin{aligned} Q_l &\sim \text{Beta}(2, 2) \\ \text{Households}_l &\sim \text{Normal}(250 + 400 Q_l, 30) \\ \text{Rent}_l &\sim \text{Normal}(500 + 1800 Q_l, 100) \\ \text{SetupCost}_l &\sim \text{Normal}(6000 (1 - Q_l), 400) \end{aligned}$$

Quality is modeled with a Beta distribution because a bounded score on $[0, 1]$ is precisely the support the Beta family occupies, and the symmetric $\text{Beta}(2, 2)$ shape encodes a defensible claim about real streets: most premises are middling, truly excellent or truly dreadful corners are rare, and no location is either mathematically perfect or entirely hopeless, since the density of a $\text{Beta}(2, 2)$ vanishes at both endpoints. A uniform distribution would instead claim that a flawless location is exactly as common as an average one, which is not a claim anyone would actually want to make.

The Normal residuals attached to rent, household count, and setup cost each collect the many small unmodeled influences on that quantity, such as building size or a landlord's temperament, and the Normal distribution is the maximum entropy choice once only a mean and a finite spread are being claimed. The economic content of the equations is worth reading separately from their distributional form. Household count is the causal bridge from geography to demand, and it is the

reason good locations command high rent: rent and foot traffic share a common cause in location quality, and neither directly causes the other. Setup cost falls as quality rises because run-down premises with cheap rent need new wiring and refrigeration, which plants a negative correlation between two cost components that a naive correlation analysis would otherwise misread as a direct relationship, when in fact both are consequences of the same underlying quality.

4.2 Customers: budgets, schedules, and tastes

Not every household in a location's catchment becomes a customer. The pool of potential regulars is drawn as $N_l \sim \text{Binomial}(\text{Households}_l, 0.6)$, which is not a convenience but the actual generative story: each household independently either adopts the store into its routine or does not, at a common probability, and counting independent yes or no trials is exactly the setting in which the Binomial distribution is the maximum entropy choice.

Each customer who does adopt the store is given a weekly grocery budget,

$$B_i \sim \text{LogNormal}(\log 85, 0.35),$$

so that the median customer spends eighty five euros a week with a realistic right tail of larger households. The LogNormal is the correct shape for the same reason incomes are approximately log normal in the population at large: a household's budget is the product of several factors, namely household size, income level, and consumption habits, and multiplying many small positive factors produces a LogNormal distribution by the same central limit argument that produces a Normal distribution when factors are added rather than multiplied. The parameterization is also directly interpretable, since e^μ is exactly the median and the scale $\sigma = 0.35$ places roughly ninety five percent of customers between forty two and one hundred seventy euros a week.

Shopping rhythm is modeled with a primary day d_i drawn from a categorical distribution weighted toward the weekend, together with two further probabilities,

$$\pi_i \sim \text{Beta}(9, 1.5), \quad p_i \sim \text{Beta}(1.5, 12),$$

where π_i is the probability of actually shopping on the primary day (schedule adherence) and p_i is the probability of an extra, midweek top up visit on any other day. The first piles its mass near one, so that habits mostly hold but a minority of customers remain erratic; the second piles its mass near zero, so that most days bring no extra visit but the possibility is never exactly zero. In both cases the Beta distribution is required because the object being modeled is a probability, and in both cases the chosen parameters keep the density away from the endpoints, so that no customer in the model is a perfectly deterministic machine.

Tastes are represented in three parts. A weekly basket composition across the twelve merchandise categories, $w_i \sim \text{Dirichlet}(\alpha)$, is the natural generalization of the Beta to a set of

shares that must sum to one, and its concentration vector α simultaneously fixes the population's average basket, through the relative size of its entries, and controls how much individual customers deviate from that average, through its overall magnitude. Price sensitivity is drawn from $\beta_i \sim \text{LogNormal}(0, 0.4)$, which must be strictly positive because every demand curve in the model slopes downward, and whose median of one anchors the scale of the entire utility function introduced in the next subsection. Brand affinity, meaning the pull toward premium or budget merchandise, is deliberately constructed to correlate with budget:

$$b_i = \text{logistic}(0.6 z_i + 0.8 \nu_i), \quad \nu_i \sim \text{Normal}(0, 1),$$

where z_i is the customer's standardized log budget and ν_i is fresh, independent noise. This is a standard latent Gaussian device in which an unbounded latent trait is built with a controlled correlation and then squashed into the unit interval with a logistic transform, and the two loadings are chosen so that $0.6^2 + 0.8^2 = 1$, which keeps the latent score on a standardized scale and makes the loading on budget, 0.6, exactly equal to the correlation between brand affinity and budget by construction. The correlation itself is required because in reality wealthier households do lean toward premium brands, and a dataset that failed to reproduce it would fail a basic realism check before any analysis began.

Finally, payment behavior is built from a persistent type rather than a single flat probability, because a flat rate would make payment method pure noise, uninformative about who is standing at the till:

$$T_i \sim \text{Bernoulli}(0.6), \quad \text{PaysByCard}_{iv} \sim \text{Bernoulli}(0.95 T_i + 0.10 (1 - T_i)),$$

so that a card preferring customer, $T_i = 1$, pays by card on ninety five percent of visits and a cash preferring customer on only ten percent. Card payments carry the customer's identity forward into the receipts; cash payments do not, which is the single mechanism that gives the eventual dataset its realistic, partial customer level visibility.

4.3 The discrete choice model

The single most load bearing structure in the entire model is the rule by which a customer, standing before a shelf of competing products within one merchandise category, decides what to buy. Each product s offers the customer a utility

$$U_{is} = \theta_{ic} + a_s + \gamma (1 - |b_i - \text{BrandLevel}_s|) - \beta_i \text{Price}_s,$$

where θ_{ic} is the customer's current need for the category, γ weights the fit between the customer's brand affinity and the product's brand level, and β_i is the customer's own price sen-

sitivity. The term $a_s \sim \text{Normal}(0, 0.8)$ is an unmodeled per-product appeal intercept, standing in for the thousand unmodeled reasons one product outsells its shelf neighbor, such as packaging or familiarity; a Normal is the maximum entropy choice given only a scale, and this particular scale is wide enough that a few products in every category sell so little all year that they amount to realistic dead stock, a fixture of every real assortment. Alongside every stocked product sits an outside option, representing the choice to buy nothing here or to buy elsewhere, and the customer’s actual choice is drawn as $\text{choice} \sim \text{Categorical}(\text{softmax}(U_{is_1}, \dots, U_{is_K}, U_0))$.

The softmax form is not a convenient approximation; it is the exact consequence of a well known result in discrete choice theory. If each product’s utility is disturbed by an independent, identically distributed Gumbel shock and the customer simply picks the option with the highest noisy utility, the resulting choice probabilities are exactly the softmax of the mean utilities. This is McFadden’s random utility result (McFadden, 1974), and it means the smooth formula above and the human story of a shopper with genuine but noisy tastes are the same model described two ways. It also means the model the customer is implicitly following is exactly the model a real discrete choice analyst would fit to purchase histories, so that grading a later analytical exercise against the true parameters is a fair comparison of like against like, rather than a comparison against a mechanism the analyst was never given a chance to recover.

This single structure produces, without any additional machinery, four properties that a realistic market must have. Demand slopes downward, because raising a product’s price lowers its utility and therefore its choice probability, at a rate set by each customer’s own price sensitivity. Substitution follows a sensible pattern, because a price increase on one product pushes probability mass onto its nearest neighbors in brand level, not onto arbitrary alternatives. Budget discipline emerges naturally, because a customer works through a shopping list item by item until the budget binds. Hidden demand is recorded automatically, because when a customer’s first choice is out of stock, the original choice is logged as a latent truth before the customer re-evaluates the softmax over what remains, giving an honest ledger of demand that never appears in the sales data.

4.4 Owner beliefs and the cost of opening

The entrepreneur begins with a fixed sum of capital and three standing policies inherited from ordinary small business practice rather than optimized from scratch: fixed opening hours; a fixed markup m_c applied uniformly within each merchandise category, so that a product’s shelf price is set mechanically as $\text{Price}_s = \text{charm}((1 + m_c) \text{Cost}_s)$, with the charm function rounding to the SKU’s own habitual psychological ending, such as a price ending in nine cents; and a flat safety stock rule requiring a constant fraction $\eta = 0.30$ of a month’s believed category demand to sit on the shelf at all times. A fourth belief, that no single product should capture more than a share $\rho = 0.05$ of its category’s demand, rounds out the policies that the opening decision below

will actually optimize under.

Before any belief can be formed there is a fact to be believed about. The true demand potential of a location for a merchandise category, the quantity the whole opening decision ultimately turns on, is defined simply as

$$D_{cl} = \text{Households}_l \cdot \mu_c,$$

where μ_c is the mean monthly per household consumption of category c that a store of this kind could realistically capture, already net of the households who never adopt the store at all and of the share of spending a neighborhood shop can expect to win against larger competitors. This quantity is ground truth: nobody in the model, least of all the owner, observes it directly.

The owner cannot observe the true demand potential of a location, so his belief is modeled as the truth seen through a noisy and mildly optimistic lens, $\hat{D}_{cl} = D_{cl} e^{\delta_c}$ with $\delta_c \sim \text{Normal}(0.10, 0.25)$. The error is placed on the logarithmic scale because people misjudge quantities in ratios rather than in absolute units; nobody thinks their estimate is off by thirty seven boxes, they think demand is probably a fifth higher or lower than it looks. Exponentiating a Normal error produces a belief that is LogNormal around the truth, automatically positive as a demand estimate must be, and the positive mean of δ_c encodes a documented behavioral regularity, namely that new business owners tend to be mildly overoptimistic. This single design choice is what causes the store to systematically overstock in its early period, and the size of that overstocking can be computed exactly from the belief error rather than being asserted as an unexplained fact about the data.

The cost of opening the store is written as an explicit structural causal model, meaning each cost component is caused by, rather than merely equal to, the decisions and location features that produce it:

$$\begin{aligned} \text{TotalCost}_0 &\stackrel{\leftarrow}{=} \text{FixedCost}_0 + \text{VariableCost}_0 \\ \text{FixedCost}_0 &\stackrel{\leftarrow}{=} \text{Rent}_0 + \text{SetupCost}_0 \\ \text{VariableCost}_0 &\stackrel{\leftarrow}{=} \text{RestockCost}_0 + \text{StorageCost}_0 \\ &\quad + \text{OverheadCost}_0 + \text{UtilityCost}_0 + \text{ListingCost}_0 \end{aligned}$$

One detail deserves particular emphasis because it is a solvency condition rather than a stylistic one: the owner himself draws no wage in this accounting, since a sole proprietor is paid out of profit, not out of payroll. A numerical audit of the model confirmed that charging the owner a market wage for his own long working days makes every affordable opening plan show a believed loss, which no rational entrepreneur would actually sign, so leaving the owner unpaid is required for the model to describe a decision anyone would make.

4.5 The opening decision as a mixed integer program

With beliefs, costs, and policies fixed, the owner's opening decision is formulated as a mixed integer linear program, choosing simultaneously which location to occupy, which products to list, how much of each to buy, and how many additional staff to hire, so as to maximize the profit he believes he will earn, subject to the constraints he actually knows about. Five decision variables carry the whole problem: $y_l \in \{0, 1\}$ marks which single location is chosen, $x_s \in \{0, 1\}$ marks which products are listed on the shelf at all, $q_s \geq 0$ is the quantity of product s purchased at opening, $u_s \geq 0$ is the quantity the owner believes he will sell of it before the objective ever needs the true answer, and $E_0^h \geq 0$ is the number of additional staff hired beyond the owner himself.

$$\max \sum_c \sum_{s \in S_c} (1 + m_c) \text{Cost}_s \cdot u_s - \text{TotalCost}_0$$

subject to

$$\begin{aligned} \sum_l y_l &= 1 && \text{(exactly one location)} \\ u_s &\leq q_s && \text{(cannot sell what is not stocked)} \\ \sum_{s \in S_c} u_s &\leq \sum_l y_l \hat{D}_{cl} && \text{(believed category demand)} \\ u_s, q_s &\leq \rho \sum_l y_l \hat{D}_{cl} && \text{(no product dominates its category)} \\ \sum_{s \in S_c} q_s &\geq \eta \sum_l y_l \hat{D}_{cl} && \text{(the safety stock rule)} \\ \sum_s q_s &\leq \sum_l y_l \text{ShelfCapacity}_l && \text{(shelf capacity)} \\ \sum_s x_s &\leq \sum_l y_l \text{ShelfSlots}_l && \text{(shelf slots)} \\ E_0^h &\geq \sum_l y_l \text{OperationalNeeds}_l - 1 && \text{(minimum staffing)} \\ \text{TotalCost}_0 &\leq \text{InitialCapital}_0 && \text{(capital)} \end{aligned}$$

Here $\rho = 0.05$ is the owner's flat belief that no single product should capture more than a twentieth of its category's demand, and $\eta = 0.30$ is the fraction of a month's believed category demand he insists on holding as safety stock at all times, the same two policy constants introduced at the start of this subsection.

A mixed integer linear program is the natural formulation here, rather than a simpler heuristic, because the decision genuinely is a joint discrete and continuous optimization: the location choice and the listing decision for each product are binary, the purchase quantities are continuous,

and every constraint linking them, such as capital and shelf space, is linear once the location is fixed. Solving the joint problem rather than comparing separate solutions for each candidate location keeps the model honest about the fact that a real entrepreneur weighs locations against each other using the very same capital and the very same assortment logic, not eight independent plans evaluated in isolation.

This formulation produces several properties without any of them being separately engineered. Location choice becomes a genuine trade off between rent and believed demand, exactly as a real entrepreneur weighs traffic against cost. Product variety emerges from the interaction of beliefs, capital, and the no domination constraint, rather than being a hand set number: the breadth of the assortment is an equilibrium outcome of the optimization, not a parameter. And because the belief error introduced above carries a positive mean, the optimal solution systematically overstocks relative to the truth, so that when the analyst later finds bloated inventory in the cost sheets, there is a genuine mechanism behind it rather than an injected anomaly.

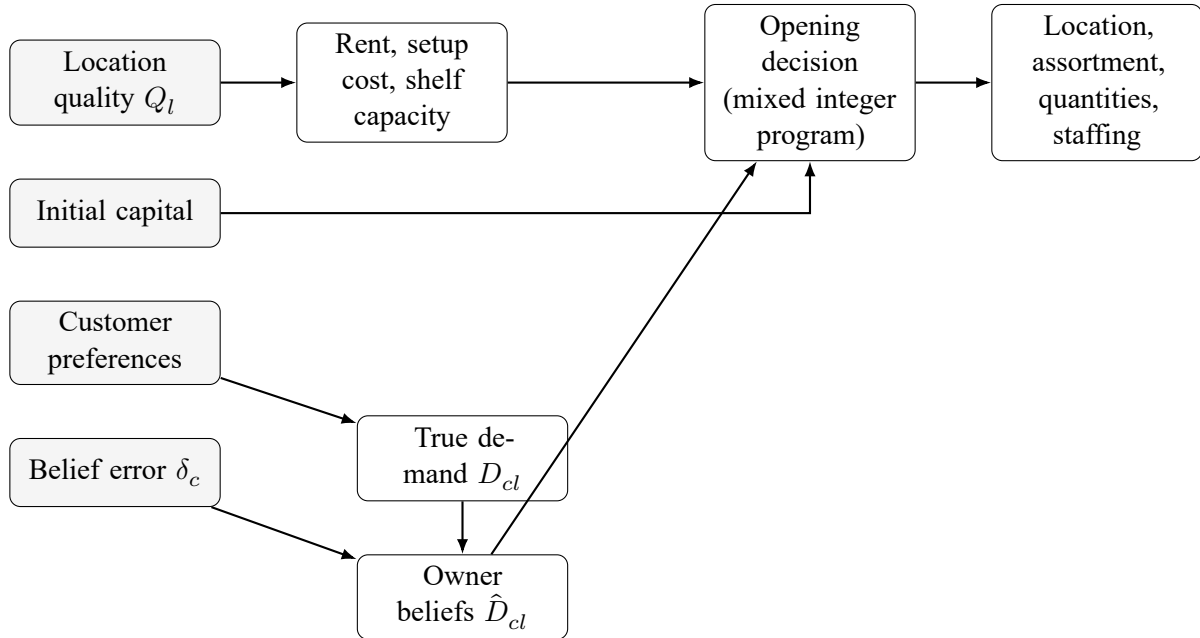


Figure 2: *The causal skeleton of Phase One. Everything upstream of the decision node is generative ground truth; the decision node and its consequences are the owner’s boundedly rational response to it.*

4.6 Why the owner is wrong, and why that is the point

Figure 2 summarizes the causal structure of Phase One. Every arrow upstream of the decision node describes objective features of the world; the decision node and everything downstream of it describes the owner’s response to a distorted and simplified picture of that world. Five specific divergences between belief and truth are built into this phase, and each is a documented, discoverable fact rather than an arbitrary flaw: optimistic demand beliefs, a flat safety stock rule

applied identically regardless of season, category level markups that ignore individual price sensitivity, static beliefs that never anticipate the seasonal cycle introduced in the next phase, and a flat product share cap that ignores genuinely heterogeneous tastes. Three profit figures eventually exist side by side once the later phases are complete: what the owner believed he would earn, what he actually earns, and what a perfectly informed operator could have earned given the same world. The distances between these three numbers are the quantified value of better information and better decisions, and producing that comparison honestly, rather than asserting it, is arguably the central purpose of the entire construction.

5 Phase Two: The Exogenous Script of the Year

Phase One built a world frozen at a single instant. Phase Two supplies its motion across the year that follows: the weather, the calendar, macroeconomic shocks, and idiosyncratic household budget shocks, all decided in advance and independent of anything the simulated agents subsequently do. This independence is not a simplification made for convenience; it is what licenses treating weather as a natural experiment and cost shocks as instruments in later analysis, because nothing about the shop’s own behavior could possibly have caused the weather.

5.1 Calendar and weather

The simulated year runs three hundred sixty five days, aligned to a real calendar year so that seasons, holidays, and an analyst’s ordinary intuitions all line up, including two days on which the store is legally closed.

Temperature is modeled as a deterministic annual cycle plus a persistent stochastic anomaly, $T_t = \mu_T(t) + \varepsilon_t$, with $\mu_T(t) = 12 + 9 \cos\left(\frac{2\pi(t-200)}{365}\right)$ and the anomaly following a first order autoregressive process, $\varepsilon_t = \phi \varepsilon_{t-1} + \text{Normal}(0, \sigma_\varepsilon)$ with $\phi = 0.7$. The autoregressive form is chosen because it is the maximum entropy stationary process consistent with exactly two facts about weather that are worth encoding and no more: it has a stable typical spread around the seasonal mean, and warm or cold spells persist for several days rather than resetting independently each morning. The decomposition into a deterministic cycle and a stochastic anomaly is not a cosmetic convenience either; it is the causal design that allows later analysis to distinguish the predictable effect of season from the genuinely surprising effect of weather, since the anomaly is constructed to be orthogonal to the calendar by design, while in raw data the two are thoroughly entangled because July is warm precisely because it is July.

Rainfall requires two separate decisions, whether it rains at all and how much, and each has its own natural distribution. Occurrence follows a two state Markov chain rather than an independent coin flip at each day,

$$W_t \mid W_{t-1} \sim \text{Bernoulli}(p_{01}(1 - W_{t-1}) + p_{11} W_{t-1}), \quad p_{01} = 0.25, \quad p_{11} = 0.55,$$

because weather fronts pass through and wet days genuinely cluster together, which an independent process cannot represent; these transition probabilities imply a stationary wet day frequency of about thirty six percent and wet spells averaging just over two days, both ordinary for a temperate climate. Given that it rains, the amount is drawn from a Gamma distribution, $R_t \mid W_t=1 \sim \text{Gamma}(1.4, \text{mean} = 5 \text{ mm})$, the standard meteorological choice and the maximum entropy distribution on the positive real line once a mean and a mean logarithm are fixed, producing the realistic pattern of many light drizzles and few genuine downpours.

5.2 Category demand modifiers

The central object of this phase is a daily, per category demand modifier M_{ct} that scales how strongly the neighborhood wants a given category of merchandise on a given day. It is built log additively from three separate causal channels:

$$\log M_{ct} = a_c \cos\left(\frac{2\pi(t - 200)}{365}\right) + \kappa_c z_t + h_c H_t,$$

where z_t is the standardized temperature anomaly and H_t flags the days immediately preceding a major holiday. The three terms are genuinely distinct causal stories rather than three pieces of one effect. The seasonal amplitude a_c captures the predictable annual cycle, everything that makes July differ from January whether or not the weather itself cooperates, such as daylight and school terms. The anomaly loading κ_c captures the pure weather effect, an unseasonably warm week moving demand right now regardless of what the calendar says. The holiday loading h_c captures the stock up surge in the days before a major holiday. Because the model is constructed multiplicatively and expressed additively on the logarithmic scale, every loading has a clean, elasticity like interpretation, and because the temperature anomaly is orthogonal to the seasonal cycle by construction, season and weather are separately identifiable in the ground truth, which lets a later diagnostic exercise distinguish a genuine weather effect from a confounded seasonal one.

One category is deliberately given all zero loadings, meaning nothing in preferences makes it more or less wanted at any time of year. Its observed sales are nevertheless not perfectly flat in the resulting data, because every category draws from the same household budget, and when other categories surge seasonally they crowd the flat category out of a tight weekly budget. This produces a subtler lesson than a simple negative control: an analyst who finds an apparent seasonal pattern in this category and stops there has mistaken a budget mediation effect for a genuine preference effect, and only tracing the mechanism reveals the difference.

A single loading per category would, however, average away the most recognizable seasonal pattern in retail, namely that demand for products such as ice cream is dramatically higher in summer, a pattern that would otherwise be diluted by averaging it together with other, flatter products in the same broad category. Rather than assigning a separate seasonal loading to every narrow product type, which would require an unmanageable number of essentially arbitrary parameters, the model adds a narrow, surgical refinement: a small number of specific product types whose seasonality flagrantly contradicts their category's average receive their own tilt term,

$$\log \psi_{pt} = a_p \cos \left(\frac{2\pi(t - 200)}{365} \right) + \kappa_p z_t,$$

built exactly like the category modifier above but at the finer grain of a product type p within a category, and entering the same choice model introduced in Phase One as an additional utility component, $U_{ist} = U_{is} + \log \tilde{\psi}_{p(s),t}$. In winter, a customer's need for the broader category therefore slides from the summer leaning product toward its winter leaning neighbor within the same shopping decision, without any new machinery beyond the existing discrete choice structure; every other product type keeps $\psi \equiv 1$.

5.3 Foot traffic and household budgets

Weather does not only change what people buy; it changes whether they come to the store at all. A daily traffic modifier

$$\log \Lambda_t = -0.20 W_t + 0.15 H_t + \varepsilon_t^\Lambda, \quad \varepsilon_t^\Lambda = \phi^\Lambda \varepsilon_{t-1}^\Lambda + \text{Normal}(0, \sigma_\varepsilon^\Lambda), \quad \phi^\Lambda = 0.85,$$

scales every customer's visit probability downward by about eighteen percent on a wet day and upward by about sixteen percent in the days before a major holiday, with the same autoregressive logic as the temperature anomaly so that a slow week tends to stay slow for a few days rather than resetting each morning. A subtle but important consequence follows automatically from the pantry mechanism introduced in Phase Three: because a customer's need for a category grows with time since their last purchase, a customer rained off on one day simply arrives hungrier a few days later. The transaction data therefore shows the familiar retail pattern of a dip on the rainy day followed by a rebound shortly after, purely as an emergent consequence of two mechanisms that never reference each other, with no separate rebound parameter anywhere in the model.

Household budgets are not held perfectly steady across the year either. Each week's budget is built as

$$B_{iw} = B_i \cdot e^{\xi_{iw}} \cdot s_{iw} \cdot g_{iw}, \quad \xi_{iw} \sim \text{Normal}(0, 0.12),$$

where the wobble ξ_{iw} is a small multiplicative perturbation representing ordinary week to week circumstance, on the same logarithmic scale as the budget itself, and s_{iw} and g_{iw} encode two rarer, more consequential mechanisms. A tight spell sets $s_{iw} = 0.65$ for its duration, representing a genuine financial hardship such as a job loss or an unexpected repair; it begins with probability $p_w = 0.004(1 + 2\bar{g}_w)$ per week, where $\bar{g}_w \in [0, 1]$ is the scripted energy crisis's own trajectory described below, and persists for Geometric(1/6) further weeks, a memoryless duration averaging six weeks. A splurge sets $g_{iw} = 1.5$ for a single week with probability 0.01, representing a windfall or a celebration. The tight spell's entry rate is deliberately coupled to the crisis trajectory, tripling at the crisis's peak, because a real energy crisis squeezes household heating bills at the same time as it squeezes a retailer's refrigeration costs, and a model that raised the store's costs without touching any household's budget would be describing an implausible economy.

5.4 Cost shocks as a marked point process

Beneath the everyday weather sits a slower moving layer of macroeconomic shocks affecting wholesale costs, and underneath that a boring, continuous inflation drift, $\text{infl}(t) = e^{\pi t/365}$ with $\pi = 0.025$ a year, riding every cost and rate in the model. The shocks proper are modeled as a marked point process: shock events arrive according to a Poisson process,

$$K \sim \text{Poisson}(1.5), \quad \tau_k \sim \text{Uniform}(1, 365) \text{ for each event } k,$$

since the Poisson count is the maximum entropy distribution for the number of rare, independent events given only a rate, and a well known property of the Poisson process guarantees that, given the count, the arrival times are conditionally uniform across the year, which is a theorem rather than an additional assumption. One large event, an autumn energy crisis, is scripted rather than drawn at random, ensuring that every run of the model tells a genuinely interesting supply side story rather than leaving open the possibility of a year with no shocks at all.

Each event's magnitude is drawn from a LogNormal distribution,

$$\zeta_k \sim \text{LogNormal}(\log \zeta_0, 0.3),$$

because a shock's peak effect is a positive multiplicative disturbance on cost, the same support and story argument that justified every other ratio like quantity in the model, and each event's effect over time follows a deterministic shape,

$$g_k(t) = \zeta_k \cdot \min\left(1, \frac{t - \tau_k}{14}\right) \cdot \exp\left(-\frac{(t - \tau_k - 14)_+}{\rho_k}\right), \quad t \geq \tau_k,$$

ramping up linearly over two weeks and then decaying exponentially with time constant ρ_k , so that a shock is neither a permanent level change nor a transient blip but a slow wave that a forecaster genuinely has to track. Every affected category's wholesale cost multiplier then assembles the drift and every live shock log additively,

$$\log \text{CostMult}_{ct} = \frac{\pi t}{365} + \sum_k g_k(t) \mathbb{1}[c \in A_k],$$

with A_k the set of categories the event touches.

The mechanism by which these shocks reach the shelf is one of the more consequential design decisions in the whole model. The owner never observes CostMult_{ct} directly; he sees only the invoices his supplier sends him, and he tracks an exponentially smoothed trend in those invoice costs,

$$\overline{\text{Cost}}_{s,t} = \alpha \text{Cost}_{s,t} + (1 - \alpha) \overline{\text{Cost}}_{s,t-}, \quad \alpha = 0.35,$$

where the smoothing weight $\alpha = 0.35$ is a purely technical constant unrelated to the basket concentration vector of the same name in Section 4. The owner moves the shelf tag only once the smoothed cost has drifted more than three percent from the current price, a pattern known in the economics literature as menu cost pricing. The consequence is that shelf prices now vary over time for reasons entirely unconnected to the demand side of the market: a foreign harvest failure or a port strike moves costs without touching any household's budget, which is exactly the exogenous price variation an econometrician needs in order to estimate a genuine price elasticity of demand rather than a number contaminated by reverse causation from demand to price. The scripted energy crisis is a deliberate exception to this cleanliness, since it also squeezes household budgets through the tight spell mechanism described above, making it a subtly invalid instrument whose bias can, in principle, be computed exactly, turning instrument validity from an assumption asserted in a single sentence into a testable claim with a known answer.

5.5 Why exogeneity licenses counterfactual reasoning

Because Phase Two's script is written entirely in advance and never listens to anything the simulated agents do, a counterfactual world can be produced almost for free: simply edit the script, for instance deleting the scripted energy crisis, and replay the identical daily market mechanism of Phase Three with every other random draw held fixed by the keyed randomness discipline described in Section 3. The difference between the original run and the edited run is then the causal effect of the edit itself, not merely an estimate of it. This is the property that Phase Four, described in Section 7, generalizes into a full policy laboratory.

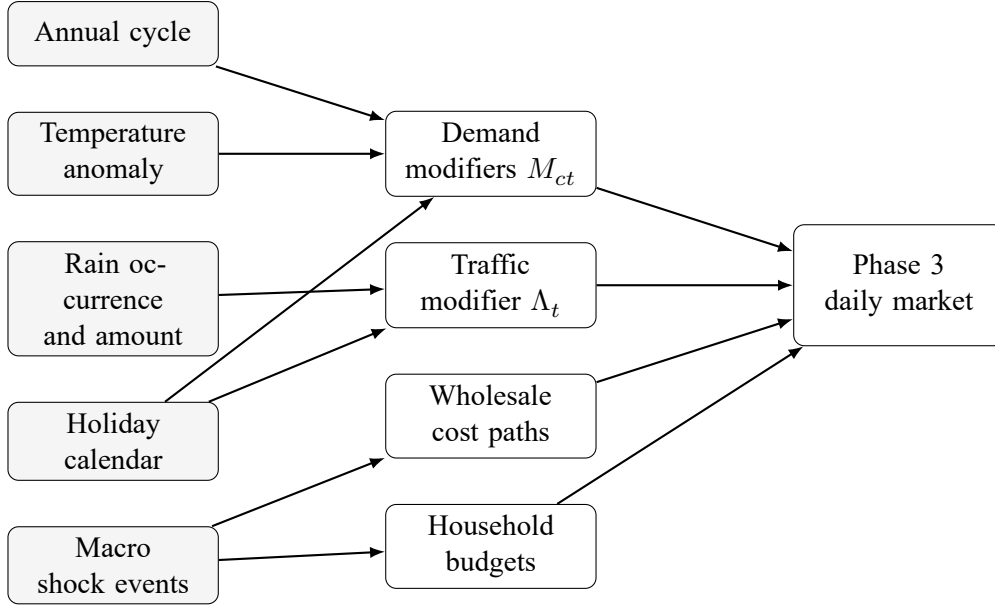


Figure 3: *The causal skeleton of Phase Two. Every arrow points into the market that Phase Three performs; nothing points back, which is what licenses treating these channels as exogenous instruments and natural experiments.*

6 Phase Three: The Daily Market Loop

Phase Three is where the two sides of the market, built and scripted in the preceding phases, finally meet. It is the only phase that produces data in the ordinary sense: a customer with a specific need walks past a shelf whose price was set according to the owner’s rule, on a day whose weather was drawn in Phase Two, and either buys, substitutes, or leaves, and every one of these outcomes either does or pointedly does not leave a trace in the records the analyst eventually receives.

6.1 The pantry and the mechanics of need

Phase One promised that a customer’s need for a category would be driven by their basket preferences and the time since their last purchase. Phase Three makes that promise mechanical through the simplest object that behaves correctly: a pantry. Each customer is given a latent stock of each category, measured in units, which drains at a steady daily rate and refills only when the customer actually buys more:

$$P_{ic,t+1} \stackrel{\leftarrow}{=} \max(0, P_{ict} - r_{ict}) + (\text{units purchased on day } t), \quad r_{ict} = r_{ic} \cdot M_{ct}.$$

Scaling the drain rate itself by the demand modifier, rather than adding the modifier directly to a customer’s momentary desire, is a deliberate refinement of the interface promised in Phase

Two, because it changes not only how much a customer wants a category but when that want actually arrives: a hot week does not make a household want ice cream in the abstract, it makes the household eat the ice cream in the freezer faster, so the container runs out sooner and the shopping trip that replaces it arrives earlier. This single change is also what makes the rain rebound described in the previous section emerge with a realistic magnitude, since a missed shopping trip simply leaves the drain running against an unreplenished pantry.

A customer's need for a category is then defined as the shortfall of their current pantry stock against a target level representing several days of comfortable cover,

$$\theta_{ict} = \alpha \frac{\max(0, P_{ic}^{\text{tgt}} - P_{ict})}{P_{ic}^{\text{tgt}}}, \quad P_{ic}^{\text{tgt}} = \tau r_{ic}, \quad \tau = 10 \text{ days},$$

with the shape parameter α , again a purely local constant unrelated to the two earlier uses of the same letter, calibrated so that a fully empty pantry presses harder on the choice model than an ordinary price difference but not harder than an extreme one. This shortfall enters the utility function of Phase One's discrete choice model directly, exactly as θ_{ic} did there. An empty pantry presses hard on the customer's decisions; a full one leaves the category to the outside option entirely, meaning the customer simply does not shop for it this trip. One bookkeeping rule is essential to the model's long run stability: when the outside option wins, meaning the customer effectively buys the item elsewhere or does without, the pantry is nonetheless refilled, with the event recorded in a hidden ledger under its own cause. Without this rule, demand lost to competitors would accumulate without bound and every basket would eventually list every category; with it, the pantry remains stationary and the store's captured share of total demand is exactly what the discrete choice model implies it should be.

6.2 Building the basket and the hidden demand ledger

A visiting customer builds a shopping list of every category whose pantry cover has fallen below a replenishment threshold, walks the list in descending order of urgency, and at each category faces the same discrete choice model introduced in Phase One, now evaluated only over products currently in stock, with any live promotional discount folding directly into the effective price the customer sees. Because a discount is treated simply as a lower price, promotional response requires no separate machinery and automatically inherits each customer's own individual price sensitivity. Having chosen a product, the customer buys toward the shortfall rather than filling it exactly,

$$\text{Quantity} = \left\lceil (P_{ic}^{\text{tgt}} - P_{ict}) \kappa \right\rceil, \quad \kappa \sim \text{Uniform}(0.7, 1.0),$$

capped by a per trip carrying limit drawn uniformly between eight and fifteen units and by whatever remains of the week's budget, so that quantity is a direct consequence of the pantry

rather than a freshly drawn random object, which is what gives the model its realistic correlation between a longer gap since the last visit and a larger basket this time.

When a customer's chosen product is out of stock, three things happen in a fixed order. First, the original intended choice is logged as a latent truth, with its cause recorded as a stockout. Second, the customer re-evaluates the same discrete choice model with the unavailable product removed from the choice set, so that probability mass flows toward the nearest neighbors in brand level, exactly the substitution pattern the original choice model was built to produce. Third, if the re-evaluated choice lands on another available product, the sale proceeds; if it lands on the outside option instead, the store has lost a sale it will never observe in its own data. Four distinct causes of invisible demand accumulate in this way over the year: an arrival outside opening hours, a stockout, an exhausted weekly budget partway through the shopping list, and the outside option simply winning on price or availability. This hidden demand ledger is the honest, mechanistic version of the claim that sales data are demand data censored in several distinct ways, each with its own recorded cause, rather than an unexplained gap between two numbers.

6.3 The owner's adaptive policies

Unlike the exogenous script of Phase Two, the owner in Phase Three reacts to what he actually observes, and what he observes is never the full truth. Once a week the owner places a restocking order, sized according to the simplest forecast a real shopkeeper actually runs in practice: a trailing four week average of his own recorded sales,

$$\hat{D}_{cw}^{\text{op}} \leftarrow \frac{1}{4} \sum_{j=1}^4 \text{Sales}_{c,w-j}.$$

He then orders up to a target sized for the review and lead time ahead of him, plus his standing safety stock rule,

$$\text{Target}_{sw} = \left(\frac{R+L}{7} + \eta \frac{30}{7} \right) \hat{D}_{cw}^{\text{op}} \tilde{\sigma}_{sw},$$

with $R = 7$ days between reviews, $L = 2$ days of lead time, and $\tilde{\sigma}_{sw}$ the product's recent share of its category's sales, smoothed with one pseudo unit so that no listed product is ever starved to zero. This rule carries two distinct, realistic flaws built directly into its structure. It is backward looking, so that a rising seasonal trend is always underestimated by an average of the weeks before it rose, and the month stale allocation $\tilde{\sigma}_{sw}$ starves whichever product's season is arriving while fattening the one whose season just left, even while the category total stays roughly on track. And it is blind to censoring, because the sales it averages are themselves truncated by the store's own past stockouts, which produces a self-reinforcing pattern sometimes called a censoring spiral: a product that stocks out repeatedly reports weak sales, earns a smaller

share of the next order, and consequently stocks out again. Both flaws are genuine, previously documented pathologies of naive forecasting, reproduced here as the mechanical consequence of an otherwise ordinary and defensible business rule, rather than injected as an arbitrary model defect.

The owner also runs two distinct kinds of promotion, and the contrast between them is itself an intended lesson. A markdown campaign is triggered endogenously whenever a category's weeks of cover on hand exceeds a threshold, targeting specifically the slowest selling third of that category's products. Because the trigger is overstock, and overstock is correlated with weak underlying demand, any naive before and after comparison of a markdown's effect is contaminated by exactly the same selection problem that afflicts real world marketing analytics, understating the campaign's true effect because the campaign was deliberately aimed at products already trending downward. A separate, calendar triggered storewide discount on the last Sunday of every month is, by contrast, entirely exogenous, announced in advance and correlated with nothing about current stock levels, providing a clean source of price variation an analyst can use to anchor an elasticity estimate before attempting to untangle the confounded markdown campaigns.

Perishable products spoil overnight,

$$\text{Spoiled}_{st} \sim \text{Binomial}(\text{OnHand}_{st}, \min(0.35, \lambda_{c(s)}^d f_{c(s),t})), \quad \lambda_c^d = 1 - (1 - \lambda_c)^{1/7},$$

where λ_c is the category's interpretable weekly base rate, converted to a nightly rate λ_c^d , and modulated by a spoilage factor built from the same log additive logic as every other multiplier in the model,

$$\log f_{ct} = \kappa_c^T \frac{T_t - \bar{T}}{10} + \kappa_c^E \bar{g}_t + \text{Normal}(0, 0.25),$$

where $\bar{T}$ is the annual mean temperature, so the heat term is a deviation from typical conditions rather than an absolute temperature, with a heat term strongest for ambient perishables such as bread and produce, a cold chain strain term loading on the energy crisis's own trajectory $\bar{g}_t$ and strongest for frozen and dairy goods, and an idiosyncratic batch luck term standing in for the reality that one delivery travels worse than another. The thinning itself uses a nightly binomial process rather than a single weekly cull, specifically because the store's inventory snapshots are visible day by day, and a once weekly cull would stamp an artificial sawtooth pattern into every perishable product's inventory series, a periodic artifact with no physical cause that a careful analyst would immediately, and correctly, flag as evidence of a simulated rather than a found dataset.

Finally, the store's cash position is tracked explicitly, with a credit line available at a fixed interest rate whenever the monthly close would otherwise take cash below zero. Restocking

orders are themselves capped by available cash and credit headroom, so that a struggling store visibly thins its own shelves months before any trouble would show up in an annual profit figure, exactly as a real distressed retailer’s inventory often signals difficulty well before its accounts do.

6.4 The recording layer: imperfect paperwork over a perfect physical reality

An early, fully reconciled version of the generated dataset was reviewed and found, on inspection, to have one serious flaw: its records were perfect. Revenue tied to the cent, every inventory identity closed exactly, and no document was ever missing, duplicated, or mistyped. Real book-keeping is never like this, and a dataset that reconciled perfectly would itself be evidence of fabrication to any experienced reader. The model therefore adds a deliberate recording layer, governed by one guiding principle: the underlying physical reality of the simulation remains untouched and internally perfect, and only the documents describing that reality are allowed to err, with every imperfection injected at the point of export rather than into the simulation’s own state.

A small number of realistic paperwork defects are injected at low, carefully calibrated rates, including duplicated till receipts from a point of sale terminal retrying an upload, invoices posted twice by a distracted clerk, occasional missing invoice lines for goods that were paid for but never logged, unlogged spoilage that only surfaces later as an unexplained monthly shrinkage, and the occasional mistyped inventory count that self corrects the following night. Each defect is drawn from its own independently keyed random stream and is simultaneously written to a separate answer key recording exactly what was changed, so that the exercise of reconciling the messy records back to the truth is itself gradable against a known, recorded answer, and the underlying simulated reality that the counterfactual machinery of Phase Four depends on is never disturbed by paperwork noise that exists only in the documents an analyst receives.

The paperwork drift this produces has its own exact algebra. Let D_{st} denote SKU s ’s cumulative book-minus-true discrepancy on day t , pushed upward by duplicate invoices and unrecorded spoilage and pushed downward by missing invoices and duplicate receipts. The inventory figure an analyst actually sees is book stock rather than the simulation’s true stock S_{st} ,

$$B_{st} = S_{st} + D_{st} - \sum_{c \leq t} A_{sc},$$

where A_{sc} is the cumulative correction posted at each monthly count. On the last day of every month the owner counts the shelf and posts the adjustment $A_{sc} = (B - S)_{sc-}$ under the write-off reason `stock_count`, snapping the book back to physical truth before it begins drifting anew; every other write-off keeps the reason `spoilage`. Because unrecorded spoilage is

the dominant contributor to D_{st} , these monthly adjustments are themselves mostly losses, reproducing the familiar shape of retail shrinkage, with occasional gains where a missing invoice or a duplicated receipt happened to tip the balance the other way. Genuine refunds are deliberately kept out of this recording layer and modeled instead as a true simulation event: after each receipt is finalized, an independently keyed stream determines whether one of its lines is returned,

$$\Pr(\text{refund} \mid \text{receipt}) = 0.006, \quad \text{return delay} \sim \text{Uniform}\{1, \dots, 13\} \text{ days},$$

posting as its own till transaction at the price actually paid, referencing the original receipt, and destroying the returned unit rather than restocking it, exactly as Section 9 describes.

6.5 The three profit figures

With a full year performed, three profit figures that Phase One promised finally become computable side by side: the profit the owner believed he would earn at the moment of opening, π_{believed} , read directly off the Phase One mixed integer program’s own objective value and extended by his own rules over the year; the profit the ledger actually shows after a full year of realistic behavior, π_{realized} , the number a real accountant would certify; and the profit an otherwise identical but perfectly informed operator could have earned given the same world, meaning the same customers, the same script, and the same policy shapes, but with true demand in place of the owner’s censored, backward looking forecast, π_{oracle} . The two gaps between these figures are exact differences between fully computed runs of the same mechanism rather than estimates,

$$\Delta_{\text{optimism}} = \pi_{\text{realized}} - \pi_{\text{believed}}, \quad \Delta_{\text{analytics}} = \pi_{\text{oracle}} - \pi_{\text{realized}},$$

where Δ_{optimism} measures the cost of the owner’s initial overoptimism and $\Delta_{\text{analytics}}$ measures the value of better information and better decisions, which is the single number the entire construction exists to make credible by actually computing it rather than merely asserting a plausible size for it.

7 Phase Four: The Policy Laboratory

Phases One through Three together produce one performed year of a single world. Phase Four turns that single performance into a laboratory in which the same world can be re-performed under a deliberately edited set of macro conditions, so that the causal effect of a policy change or an external shock can be measured directly, as an exact difference between two runs of an otherwise identical mechanism, rather than estimated statistically from a single realized history.

7.1 Scenarios as edited scripts

A scenario, in this model, is nothing more than a declarative set of edits to the exogenous script of Phase Two or to one of the owner's policy knobs, such as his opening hours or his staffing level. Concretely, a scenario spec edits one or more named quantities the earlier sections already defined: extra cost events riding the same trajectory $g_k(t)$ of Section 5, a multiplier on the daily traffic modifier Λ_t , a multiplier on the weekly household budget B_{iw} , or a new rate path $r_c(t)$ from a given date. Critically, a scenario never edits an underlying random number stream directly. Because every draw in the model is keyed by stable identity, as described in Section 3, replaying an edited script through the identical daily market mechanism reproduces every unaffected customer's decisions bit for bit, so that the entire difference between the baseline world and the edited one is attributable to the edit by construction. This is the same common random numbers discipline introduced to support the Phase Two counterfactuals, here generalized into a standing feature of the model rather than a special case. A handful of reference scenarios illustrate the range of questions this laboratory can answer, including a broad economy wide supply shock, a short severe weather event, a change in value added tax, a temporary household income transfer, and a staffing decision to hire an additional clerk and extend opening hours.

The owner's declared indifference to tax strategy is not merely asserted but is, for the profit tax specifically, provably harmless: because ProfitTax is a flat rate on a positive base, maximizing $(1 - r^\pi) \pi$ over any set of operating decisions has exactly the same maximizer as maximizing π itself,

$$\arg \max_{\text{decisions}} (1 - r^\pi) \pi(\text{decisions}) = \arg \max_{\text{decisions}} \pi(\text{decisions}),$$

so an owner who ignores the profit tax when making ordinary operating decisions loses nothing relative to one who optimizes around it. No such guarantee holds for the value added tax, whose category specific rate $r_c(t)$ changes relative prices across categories and therefore can, in principle, reward cross category arbitrage; the owner's failure to perform that arbitrage is exactly the gap a later prescriptive analysis can measure against a scenario's twin arms.

7.2 The tax layer

Because a policy laboratory without taxation would be unable to represent the most common lever through which real macroeconomic policy actually reaches individual households and businesses, this phase also introduces a complete, differentiated tax system into the baseline world itself, rather than reserving it only for edited scenarios. Value added tax is applied at a category specific rate $r_c(t)$, and the owner's shelf tag is formed from the tax exclusive, smoothed invoice cost with the current rate reapplied on top:

$$\text{Tag}_{st} = \text{charm}(\overline{\text{NetCost}}_{s,t} \cdot (1 + m_c) \cdot (1 + r_c(t))),$$

where $\overline{\text{NetCost}}_{s,t}$ is the exponentially smoothed, tax exclusive invoice cost introduced in Phase Two and m_c is the same category markup fixed in Phase One. Because the rate enters multiplicatively on the same trend the owner already tracks, a rate change reaches his *intended* price immediately and exactly, while the price customers actually see at the shelf continues to lag through the same menu cost mechanism described in Phase Two, which is precisely the kind of incomplete, delayed tax pass-through documented in the empirical public finance literature.

Value added tax is remitted monthly as the difference between output tax collected on sales and input tax paid on purchases,

$$\text{VAT}_t = \sum_c \text{Sales}_{ct} \frac{r_c}{1 + r_c} - \sum_c \text{Purchases}_{ct} \frac{r_c}{1 + r_c},$$

and this net amount leaves the till at the monthly close, the same till that funds the cash capped ordering rule of Phase Three. Payroll tax applies to any wages paid to hired staff at a flat employer rate, $\text{PayrollTax}_t = r^{\text{pay}} \text{Wages}_t$ with $r^{\text{pay}} = 0.25$, and a flat tax applies to positive annual profit, accrued in the year it is earned rather than paid immediately,

$$\text{ProfitTax} = r^\pi \cdot \max(0, \text{ProfitBeforeTax}), \quad r^\pi = 0.20.$$

The owner is modeled as tax aware but not tax strategic: he correctly prices on his net of tax margin, as the tag formula above shows, and correctly accounts for the full employer cost of hiring, meaning wages plus payroll tax, in the opening decision of Phase One, but he never performs cross category tax arbitrage or elasticity aware repricing in response to a tax change, leaving that gap as an open door for a later prescriptive analysis to identify and price.

8 Phase Five: Three Years of Growth, Churn, and Capital

A single simulated year is, in one specific and important sense, an incomplete dataset: it cannot separate a genuine growth trend from ordinary seasonality, it cannot show an individual customer arriving in or departing from the neighborhood, and its owner never faces the central question every real small business eventually faces, namely whether to reinvest, hold steady, or exit. Phase Five extends the horizon to three full years and adds exactly the three kinds of motion a single year structurally cannot contain.

8.1 The customer panel as a flow

Where earlier phases treated the customer population as fixed for the simulated year, Phase Five gives each regular customer a persistence type drawn once at creation,

$$\text{type}_i = \begin{cases} \text{rooted} & \text{with probability 0.80} \\ \text{transient} & \text{with probability 0.20,} \end{cases}$$

where a rooted customer will not leave the neighborhood within the three year window at all, and a transient customer instead faces an ongoing monthly hazard of departure, $h = 1/18$, giving a mean tenure of eighteen months. Every departure is matched, after a random delay of $1 + \text{Geometric}(0.5)$ months, by a fresh customer profile representing a new household moving into the vacated home, and a slow trickle of genuinely new arrivals, $\text{Poisson}(4)$ per year, on top of straightforward replacement represents modest organic neighborhood growth. Newly arrived customers are themselves drawn with an even fifty percent chance of either persistence type, reflecting the reasonable assumption that a household which has just moved once is more likely to move again than the settled population it replaces.

Nothing about this churn is announced anywhere in the visible data; a customer's card token simply stops appearing in the receipts one month and never returns, and a new token begins appearing elsewhere. An analyst must infer departure from silence, and must take care not to confuse a customer who has genuinely left the neighborhood with one who is merely visiting less often for other reasons, a distinction that becomes especially difficult near the very end of the observed window, where a recently silent customer cannot yet be distinguished from one who will return next month. This is a form of the right censoring problem familiar from survival analysis, and it is a genuine, unavoidable feature of any panel observed for a finite length of time, reproduced here rather than glossed over.

8.2 Retained earnings and the endogenous expansion decision

From the beginning of the second year, once the owner's books have formalized, he begins retaining a share $\rho_{RE} = 0.5$ of every profitable month's after tax result π_m as declared retained earnings, drawing the rest out as his own income,

$$\text{draw}_m = (1 - \rho_{RE}) \max(0, \pi_m), \quad \text{RE}_m = \text{RE}_{m-1} + \rho_{RE} \max(0, \pi_m) - \text{capex}_m,$$

with no draw and no retention at all in a loss month, exactly as a small business owner who has proven a concept typically begins to behave. Once retained earnings and available cash both cross fixed thresholds,

$$\text{RE}_m \geq K_{\text{exp}} \quad \text{and} \quad \text{cash}_m - \text{tax_reserve}_m \geq C_{\text{exp}},$$

the owner expands the business on the first day of the following month: he hires his first employee on a fixed part time shift, extends opening hours by two hours a day, deepens his shelf

capacity without widening the range of products carried, and pays a one time fit out cost C_{exp} from savings.

The threshold K_{exp} is calibrated, rather than tied to a fixed calendar date, so that under the realized path of the model the expansion lands in the autumn of the second year, after a scripted local population increase but before a scripted competitive entry in the following year, producing a specific and analytically important narrative: the expansion looks like an entirely sensible decision at the moment it is made, based on everything the owner could reasonably have known at the time, and only becomes questionable in hindsight once a subsequent, unrelated shock arrives. A counterfactual twin of the model, in which the expansion threshold is disabled entirely, reveals that the expansion's true cost across its first fourteen months substantially exceeds the additional margin the extra hours and capacity actually generate, because the fixed cost of an additional wage is paid in full while the additional revenue it produces is diluted by the ordinary grocery margin. This is a deliberately reproduced version of one of the most common capital allocation mistakes in small retail, namely investing capital at what retrospectively turns out to be the peak of the underlying opportunity, generated here from the interaction of an ordinary retention rule, an ordinary calibrated threshold, and an otherwise unrelated later shock, rather than scripted as an explicit narrative choice.

8.3 A world that stays unquiet

Consistent with the stated design principle that a three year history should not read as a single stable year merely repeated three times, the second and third years are populated with a specific sequence of scripted events, each chosen to test a distinct analytical skill. The second year includes an equipment failure that destroys a portion of frozen and refrigerated stock overnight, a single category cost shock affecting dairy products, a prolonged summer heatwave that simultaneously boosts demand for warm-weather products and increases spoilage pressure through the same mechanism, and the scripted population increase that helps trigger the expansion described above. The third year includes a contractual rent increase taking effect at a two year lease renewal, alongside similarly lumpy, dated steps in wages and utility tariffs, each a discrete jump on a known date rather than a smooth index, so that an analyst who treats every cost as continuously drifting inflation will misdate exactly where the squeeze came from. It also includes the entry of a discounting competitor a short distance away whose effect on individual customers is deliberately heterogeneous,

$$m_i = \exp(-\chi \cdot \tilde{s}_i \cdot k_i), \quad k_i = \begin{cases} 1.0 & \text{transient customers} \\ 0.6 & \text{rooted customers,} \end{cases}$$

a multiplicative penalty on each customer's visit probability, where m_i denotes this penalty and is unrelated to the category markup m_c of Section 4 despite the shared letter, $\tilde{s}_i$ is that

customer's percentile rank in price sensitivity, and χ is calibrated to a target aggregate visit drop, so that the competitor draws away price sensitive customers fastest, and rooted customers, held by relationships rather than price alone, are penalized at only sixty percent of the rate a transient customer at the same price sensitivity would be. This is followed by a scheduled rather than reactive pricing response by the owner to that competitor, a short seasonal street festival, and a second cost shock affecting staple goods. Two further counterfactual twins, one in which the competitor never enters and one in which the expansion never occurs, allow the true cost of each event to be measured directly rather than inferred, and a specific, intended finding of the model is that the capital decision to expand costs the business substantially more than the competitive entry does, even though the competitor is the more visible and more emotionally salient event to an owner living through it.

9 The Accounting Layer: Reconciling Money and Goods

Running beneath every phase described above is a strict accounting discipline governing how money and physical goods move between the artifacts an analyst eventually receives. One principle governs all of it: the owner's own monthly ledger is always correct about money, and every other document, particularly the raw till receipts and the raw supplier invoices, is the analyst's paperwork to reconcile against it, complete with the deliberate imperfections described in Section 6.

Revenue is recognized on a cash basis at the moment of sale, with the effective price paid reflecting, in order, the current shelf tag, any live promotional markdown, and the additional monthly storewide discount when it applies. A small share of receipts contain a voided mis-ring, in which a cashier corrects an immediate scanning error, netting to zero money and zero goods, and are distinguishable from genuine refunds by the absence of a reference to an earlier receipt. Genuine refunds, occurring at a low but nonzero rate, post as an independent till transaction referencing the original sale, reduce both cash and revenue on the day of the return, and destroy rather than restock the returned item, since a small grocer does not typically resell a returned perishable good.

Inventory is tracked as book stock rather than a continuous physical count, governed by a strict identity in which each day's change in recorded stock equals recorded deliveries minus recorded sales minus recorded write-offs,

$$\text{book}(t) - \text{book}(t - 1) = \text{deliveries}(t) - \text{sales}(t) - \text{write_offs}(t),$$

with any deviation from this identity traceable either to the deliberate snapshot typos of the recording layer or, once a month, to a physical stock count that reconciles the books and posts the difference as a shrinkage adjustment. Perishable spoilage and monthly stock count

corrections are kept in strictly separate categories precisely because they measure two different things, namely genuine physical waste and pure bookkeeping drift, and conflating them would double count the same euros.

The monthly ledger itself aggregates revenue, the cost of goods procured, rent, wages, utilities, storage, promotional costs, net tax remittance, and credit interest into a single bottom line, with a credit facility drawn automatically whenever the month's closing cash position would otherwise fall below zero. At the annual horizon this bottom line is made explicit as

$$\begin{aligned} \text{ProfitBeforeTax} \triangleq & \sum \text{Revenue} - \sum (\text{Procurement} + \text{Rent} + \text{Wages}) \\ & - \sum (\text{PayrollTax} + \text{Utilities} + \text{Storage}) \\ & - \sum (\text{Flyers} + \text{VAT} + \text{CreditInterest}) - \text{Openings}, \end{aligned}$$

where Openings collects any one time capital expenditure, such as a second location's fit-out cost, incurred that year. Profit tax is then levied only on a genuine surplus,

$$\begin{aligned} \text{ProfitTax} &= 0.20 \times \max(0, \text{ProfitBeforeTax}), \\ \text{ProfitAfterTax} &= \text{ProfitBeforeTax} - \text{ProfitTax}, \end{aligned}$$

so that a loss year triggers no tax liability but also earns no rebate, mirroring the asymmetric treatment a real small proprietorship faces. On the three year horizon this ledger additionally tracks retained earnings, owner withdrawals, one time capital expenditures, and the cash payment of the prior year's profit tax, exactly the additional bookkeeping that the expansion decision described in Section 8 depends upon.

10 Discussion

Several design choices recur across every phase of this model and are worth drawing out explicitly, because they explain why the architecture as a whole looks the way it does rather than some equally plausible alternative.

The first recurring choice is that every random variable is defended on its own terms by the same three question discipline introduced in Section 3, rather than by convention or convenience. This has a practical consequence beyond intellectual tidiness: because each distributional choice is defended by an explicit generative story, the size and shape of every planted pattern in the resulting data, such as the degree of overstocking caused by owner optimism or the seasonal swing in a particular product's sales, can in principle be computed exactly from the model's own parameters rather than merely observed empirically after the fact.

The second recurring choice is the strict separation between what is exogenous, meaning decided in advance and independent of the simulated agents' behavior, and what is endogenous, meaning a genuine response to observed outcomes within the model. Weather, macroeconomic shocks, and household budget shocks are all exogenous by construction, which is what allows them to function as instruments and natural experiments in later analysis. The owner's restocking rule, pricing rule, and expansion decision are all endogenous responses to what he actually observes, which is what allows his documented mistakes to be genuine, traceable consequences of otherwise reasonable behavior rather than arbitrary flaws inserted for pedagogical convenience.

The third recurring choice is the strict separation between the underlying physical reality of the simulated world, which is always internally consistent and exactly conserved, and the paperwork describing that reality, which is allowed to be imperfect in the same limited, realistic ways real business paperwork is imperfect. This separation is what allows the counterfactual machinery of Phases Four and Five to remain exact even in the presence of realistic recording noise, because that noise is injected only at the point of export and never touches the simulated state itself.

Finally, throughout the model a small number of specific, deliberate gaps between belief and truth, whether the owner's demand estimate, his forecasting rule, his tax strategy, or his capital allocation decision, are each introduced once, defended on realistic grounds, and then left undisturbed rather than corrected. This is a deliberate design stance rather than an oversight: the entire point of the construction is to produce a world in which the distance between what an agent believes and what is true can be measured precisely, and that distance can only be measured if the model resists the temptation to make its own protagonist smarter than the premise requires.

11 Conclusion

This paper has consolidated the theoretical reasoning behind a five phase construction of a synthetic grocery store, from the owner's opening decision solved as a mixed integer program, through a scripted year of weather and macroeconomic shocks, a daily market mechanism built on individually rational discrete choice, a laboratory of counterfactual scenarios, and a three year extension adding customer turnover and capital accumulation. Every distributional and structural choice throughout the model was defended by the same discipline: identifying the variable's true support, articulating the generative story that actually produces it, and selecting, within those constraints, the distribution that assumes the least beyond what has actually been claimed. The result is a world built from first principles rather than a spreadsheet of plausible numbers, in which every observable pattern, whether a seasonal swing in ice cream sales, an overstocked category, or a capital decision that looks reasonable at the time and questionable in hindsight, traces to an explicit, defensible mechanism rather than to an arbitrary injection of noise.

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